KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3122 | 3 Credits: 3 | Program Core | Source-grounded

Wood Chemistry

 To provide knowledge about the chemistry of raw materials in wood based industries

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3122 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3122.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Wood and Paper Technology
Course code	3122
Course title	Wood Chemistry
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L:2, T:1, P:0) Periods per semester: 45

Item	Official information
Periods per semester	45
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

14 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide knowledge about the chemistry of raw materials in wood based industries
- To familiarize students about wood analysis techniques
- To impart knowledge about the manufacture of valuable products from wood other

Course outcomes

CO	Official outcome	Study level
CO1	carbohydrates. Propose the application of physical and 9 Applying	See official syllabus
CO2	nuclear chemistry in wood Discuss the structure ,cell wall composition of 14 Applying	See official syllabus
CO3	wood and its chemical treatment Illustrate thewood hydrolysis and destructive 9 Understanding	See official syllabus
CO4	distillation of wood.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3 3 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	compounds mainly carbohydrates.	4	As specified
Module 2	Propose the application of physical &nuclear chemistry in wood.	3	As specified
Module 3	treatment	4	As specified
Module 4	Illustrate the wood hydrolysis and destructive distillation of wood.	3	As specified

MODULE 1

compounds mainly carbohydrates.

This module is presented from the official syllabus scope for Wood Chemistry. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: compounds mainly carbohydrates.
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

3 Understanding nomenclature Discuss the preparation, properties and uses of

3 Understanding nomenclature Discuss the preparation, properties and uses of is an official topic in Module 1 of Wood Chemistry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding nomenclature Discuss the preparation, properties and uses of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding nomenclature discuss the preparation, properties and uses of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding nomenclature Discuss the preparation, properties and uses of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 3 Understanding nomenclature Discuss the preparation, properties and uses of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

2 Understanding organic compounds Explain the structure of cellulose, glucose and

2 Understanding organic compounds Explain the structure of cellulose, glucose and is an official topic in Module 1 of Wood Chemistry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding organic compounds Explain the structure of cellulose, glucose and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding organic compounds explain the structure of cellulose, glucose and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding organic compounds Explain the structure of cellulose, glucose and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 1-□□ 2 Understanding organic compounds Explain the structure of cellulose, glucose and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding fructose List wood extractive terpenes and their isolation

4 Understanding fructose List wood extractive terpenes and their isolation is an official topic in Module 1 of Wood Chemistry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding fructose List wood extractive terpenes and their isolation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding fructose list wood extractive terpenes and their isolation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding fructose List wood extractive terpenes and their isolation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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but different term.

Malayalam support: Module 1-□□ 4 Understanding fructose List wood extractive terpenes and their isolation □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

2 Understanding from plants Organic Chemistry Classification of organic compounds - saturated and unsaturated compounds. IUPAC - nomenclature of simple organic compounds. Industrial methods of preparation, properties and uses of : Alcohol, phenol, aldehydes, ketones, carboxyllic acid, amines, urea, poly - functional compounds - glycol, glycerol, Carbohydrates - classification of sugars, structure of glucose, fructose and cellulose, important reaction of these compounds. Terpenes - Essential oils, classification of terpenoids, Isolation, Isoprene rule and the examples of terpenes.

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Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding from plants Organic Chemistry Classification of organic compounds - saturated and unsaturated compounds. IUPAC - nomenclature of simple organic compounds. Industrial methods of preparation, properties and uses of : Alcohol, phenol, aldehydes, ketones, carboxyllic acid, amines, urea, poly - functional compounds - glycol, glycerol, Carbohydrates - classification of sugars, structure of glucose, fructose and cellulose, important reaction of these compounds. Terpenes - Essential oils, classification of terpenoids, Isolation, Isoprene rule and the examples of terpenes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding from plants organic chemistry classification of organic compounds - saturated and unsaturated compounds. iupac - nomenclature of simple organic compounds. industrial methods of preparation, properties and uses of : alcohol, phenol, aldehydes, ketones, carboxyllic acid, amines, urea, poly - functional compounds - glycol, glycerol, carbohydrates - classification of sugars, structure of glucose, fructose and cellulose, important reaction of these compounds. terpenes - essential oils, classification of terpenoids, isolation, isoprene rule and the examples of terpenes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding from plants Organic Chemistry Classification of organic compounds - saturated and unsaturated compounds. IUPAC - nomenclature of simple organic compounds. Industrial methods of preparation, properties and uses of : Alcohol, phenol, aldehydes, ketones, carboxyllic acid, amines, urea, poly - functional compounds - glycol, glycerol, Carbohydrates - classification of sugars, structure of glucose, fructose and cellulose, important reaction of these compounds. Terpenes - Essential oils, classification of terpenoids, Isolation, Isoprene rule and the examples of terpenes. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
compounds mainly carbohydrates.
             Name the organic compounds using IUPAC
  M1.01                                                               3
Understanding
             nomenclature
             Discuss the preparation, properties and uses of
  M1.02                                                               2
Understanding
             organic compounds
             Explain the structure of cellulose, glucose and
  M1.03                                                               4
Understanding
             fructose
             List wood extractive terpenes and their isolation
  M1.04                                                               2
Understanding
             from plants

 Contents:
 Organic Chemistry
 Classification of organic compounds - saturated and unsaturated compounds.
 IUPAC - nomenclature of simple organic compounds.
 Industrial methods of preparation, properties and uses of :
 Alcohol, phenol, aldehydes, ketones, carboxyllic acid, amines, urea, poly -
functional
 compounds - glycol, glycerol,
 Carbohydrates - classification of sugars, structure of glucose, fructose and
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Propose the application of physical &nuclear chemistry in wood.

This module is presented from the official syllabus scope for Wood Chemistry. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Propose the application of physical &nuclear chemistry in wood.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

3 Understanding corrosion on wood Analyse the colloids and their importance in

3 Understanding corrosion on wood Analyse the colloids and their importance in is an official topic in Module 2 of Wood Chemistry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding corrosion on wood Analyse the colloids and their importance in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding corrosion on wood analyse the colloids and their importance in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding corrosion on wood Analyse the colloids and their importance in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Understanding corrosion on wood Analyse the colloids and their importance in □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

3 Applying paper making chemistry

3 Applying paper making chemistry is an official topic in Module 2 of Wood Chemistry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying paper making chemistry using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying paper making chemistry should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying paper making chemistry means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 3 Applying paper making chemistry [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Apply the carbon dating to find the age of wood 3 Applying Physical Chemistry Corrosion-Theory of corrosion-dry & wet, methods to prevent corrosion. Degradation of wood by products of metal corrosion. Introduction to colloids, mechanical, electrical and optical properties of colloids. Gels & Emulsions; Importance of papermaking chemistry; Fiber-fiber water bonding; Rheology, surface energy, and surface tension of colloidal systems. Electro kinetic properties, Sorption and swelling of cellulosic materials. Nuclear Chemistry Natural and artificial radioactivity, radioisotopes and their uses, carbon dating in wood Discuss the structure, cell wall composition of wood and its chemical

Apply the carbon dating to find the age of wood 3 Applying Physical Chemistry Corrosion-Theory of corrosion-dry & wet, methods to prevent corrosion. Degradation of wood by products of metal corrosion. Introduction to colloids, mechanical, electrical and optical properties of colloids. Gels & Emulsions; Importance of papermaking chemistry; Fiber-fiber water bonding; Rheology, surface energy, and surface tension of colloidal systems. Electro kinetic properties, Sorption and swelling of cellulosic materials. Nuclear Chemistry Natural and artificial radioactivity, radioisotopes and their uses, carbon dating in wood Discuss the structure, cell wall composition of wood and its chemical is an official topic in Module 2 of Wood Chemistry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Apply the carbon dating to find the age of wood 3 Applying Physical Chemistry Corrosion-Theory of corrosion-dry & wet, methods to prevent corrosion. Degradation of wood by products of metal corrosion. Introduction to colloids, mechanical, electrical and optical properties of colloids. Gels & Emulsions; Importance of papermaking chemistry; Fiber-fiber water bonding; Rheology, surface energy, and surface tension of colloidal systems. Electro kinetic properties, Sorption and swelling of cellulosic materials. Nuclear Chemistry Natural and artificial radioactivity, radioisotopes and their uses, carbon dating in wood Discuss the structure, cell wall composition of wood and its chemical using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, apply the carbon dating to find the age of wood 3 applying physical chemistry corrosion-theory of corrosion-dry & wet, methods to prevent corrosion. degradation of wood by products of metal corrosion. introduction to colloids, mechanical, electrical and optical properties of colloids. gels & emulsions; importance of papermaking chemistry; fiber-fiber water bonding; rheology, surface energy, and surface tension of colloidal systems. electro kinetic properties, sorption and swelling of cellulosic materials. nuclear chemistry natural and artificial radioactivity, radioisotopes and their uses, carbon dating in wood discuss the structure, cell wall composition of wood and its chemical should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Apply the carbon dating to find the age of wood 3 Applying Physical Chemistry Corrosion-Theory of corrosion-dry & wet, methods to prevent corrosion. Degradation of wood by products of metal corrosion. Introduction to colloids, mechanical, electrical and optical properties of colloids. Gels & Emulsions; Importance of papermaking chemistry; Fiber-fiber water bonding; Rheology, surface energy, and surface tension of colloidal systems. Electro kinetic properties, Sorption and swelling of cellulosic materials. Nuclear Chemistry Natural and artificial radioactivity, radioisotopes and their uses, carbon dating in wood Discuss the structure, cell wall composition of wood and its chemical means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Apply the carbon dating to find the age of wood 3 Applying Physical Chemistry Corrosion-Theory of corrosion-dry & wet, methods to prevent corrosion. Degradation of wood by products of metal corrosion. Introduction to colloids, mechanical, electrical and optical properties of colloids.

Gels & Emulsions; Importance of papermaking chemistry; Fiber-fiber water bonding; Rheology, surface energy, and surface tension of colloidal systems. Electro kinetic properties, Sorption and swelling of cellulosic materials. Nuclear Chemistry Natural and artificial radioactivity, radioisotopes and their uses, carbon dating in wood Discuss the structure, cell wall composition of wood and its chemical □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Propose the application of physical &nuclear chemistry in wood.
             Describe the theory of corrosion and effect of
  M2.01                                                              3
Understanding
             corrosion on wood
            Analyse the colloids and their importance in
  M2.02                                                              3
Applying
            paper making chemistry

  M2.03     Apply the carbon dating to find the age of wood          3
Applying

            Series Test – I                                          1

Contents:
Physical Chemistry
Corrosion-Theory of corrosion-dry & wet, methods to prevent corrosion.
Degradation of
wood by products of metal corrosion.
Introduction to colloids, mechanical, electrical and optical properties of
colloids. Gels &
Emulsions; Importance of papermaking chemistry; Fiber-fiber water bonding;
Rheology,
surface energy, and surface tension of colloidal systems. Electro kinetic
properties,
Sorption and swelling of cellulosic materials.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

treatment

This module is presented from the official syllabus scope for Wood Chemistry. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: treatment
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Understanding between soft wood and hardwood Explain cell wall structure and composition of

3 Understanding between soft wood and hardwood Explain cell wall structure and composition of is an official topic in Module 3 of Wood Chemistry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding between soft wood and hardwood Explain cell wall structure and composition of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding between soft wood and hardwood explain cell wall structure and composition of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding between soft wood and hardwood Explain cell wall structure and composition of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 3-□□ 3 Understanding between soft wood and hardwood Explain cell wall structure and composition of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Understanding wood

3 Understanding wood is an official topic in Module 3 of Wood Chemistry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding wood using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding wood should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding wood means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 3 Understanding wood [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize extractives and their effect on wood 3 Understanding Explain the chemical treatment and effect of

Summarize extractives and their effect on wood 3 Understanding Explain the chemical treatment and effect of is an official topic in Module 3 of Wood Chemistry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize extractives and their effect on wood 3 Understanding Explain the chemical treatment and effect of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize extractives and their effect on wood 3 understanding explain the chemical treatment and effect of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize extractives and their effect on wood 3 Understanding Explain the chemical treatment and effect of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Summarize extractives and their effect on wood 3 Understanding Explain the chemical treatment and effect of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

5 Applying chemicals on wood Chemistry of Fibrous Raw Materials Structure of softwoods, hardwoods, and non-woods; Pulpwood species; Ultrastructure of cell wall, cell wall components - cellulose, hemi cellulose, lignin, extractives and inorganic components. Difference in cell wall composition of softwood and hard wood Wood extractives - Essential oils, Tannins, flavanoids and dyestuff, properties and uses. Effect of extractives on durability of wood, gluing characteristics, surface coating. Pectin - uses Chemical Treatment of wood - chemical resistance of softwood and hardwood. Effect of chemicals on wood - aqueous solutions of inorganic salts, alkaline degradation of wood, Oxidation and bleaching of wood. Action of water - effect of moisture content on strength & electrical conductivity of wood

5 Applying chemicals on wood Chemistry of Fibrous Raw Materials Structure of softwoods, hardwoods, and non-woods; Pulpwood species; Ultrastructure of cell wall, cell wall components - cellulose, hemi cellulose, lignin, extractives and inorganic components. Difference in cell wall composition of softwood and hard wood Wood extractives - Essential oils, Tannins, flavanoids and dyestuff, properties and uses. Effect of extractives on durability of wood, gluing characteristics, surface coating. Pectin - uses Chemical Treatment of wood - chemical resistance of softwood and hardwood. Effect of chemicals on wood - aqueous solutions of inorganic salts, alkaline degradation of wood, Oxidation and bleaching of wood. Action of water - effect of moisture content on strength & electrical conductivity of wood is an official topic in Module 3 of Wood Chemistry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Applying chemicals on wood Chemistry of Fibrous Raw Materials Structure of softwoods, hardwoods, and non-woods; Pulpwood species; Ultrastructure of cell wall, cell wall components - cellulose, hemi cellulose, lignin, extractives and inorganic components. Difference in cell wall composition of softwood and hard wood Wood extractives - Essential oils, Tannins, flavanoids and dyestuff, properties and uses. Effect of extractives on durability of wood, gluing characteristics, surface coating. Pectin - uses Chemical Treatment of wood - chemical resistance of softwood and hardwood. Effect of chemicals on wood -

aqueous solutions of inorganic salts, alkaline degradation of wood, Oxidation and bleaching of wood. Action of water - effect of moisture content on strength & electrical conductivity of wood using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 applying chemicals on wood chemistry of fibrous raw materials structure of softwoods, hardwoods, and non-woods; pulpwood species; ultrastructure of cell wall, cell wall components - cellulose, hemi cellulose, lignin, extractives and inorganic components. difference in cell wall composition of softwood and hard wood wood extractives - essential oils, tannins, flavanoids and dyestuff, properties and uses. effect of extractives on durability of wood, gluing characteristics, surface coating. pectin - uses chemical treatment of wood - chemical resistance of softwood and hardwood. effect of chemicals on wood - aqueous solutions of inorganic salts, alkaline degradation of wood, oxidation and bleaching of wood. action of water - effect of moisture content on strength & electrical conductivity of wood should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Applying chemicals on wood Chemistry of Fibrous Raw Materials Structure of softwoods, hardwoods, and non-woods; Pulpwood species; Ultrastructure of cell wall, cell wall components - cellulose, hemi cellulose, lignin, extractives and inorganic components. Difference in cell wall composition of softwood and hard wood Wood extractives - Essential oils, Tannins, flavanoids and dyestuff, properties and uses. Effect of extractives on durability of wood, gluing characteristics, surface coating. Pectin - uses Chemical Treatment of wood - chemical resistance of softwood and hardwood. Effect of chemicals on wood - aqueous solutions of inorganic salts, alkaline degradation of wood, Oxidation and bleaching of wood. Action of water - effect of moisture content on strength & electrical conductivity of wood means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 5 Applying chemicals on wood Chemistry of Fibrous Raw Materials Structure of softwoods, hardwoods, and non-woods; Pulpwood species; Ultrastructure of cell wall, cell wall components - cellulose, hemi cellulose, lignin, extractives and inorganic components. Difference in cell wall composition of softwood and hard wood Wood extractives - Essential oils, Tannins, flavanoids and dyestuff, properties and uses. Effect of extractives on durability of wood, gluing characteristics, surface coating. Pectin - uses Chemical Treatment of wood - chemical resistance of softwood and hardwood. Effect of chemicals on wood - aqueous solutions of inorganic salts, alkaline degradation of wood, Oxidation and bleaching of wood. Action of water - effect of moisture content on strength & electrical conductivity of wood □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
treatment
            Describe the Anatomy of wood, difference
  M3.01                                                               3
Understanding
            between soft wood and hardwood
            Explain cell wall structure and composition of
  M3.02                                                               3
Understanding
            wood
  M3.03     Summarize extractives and their effect on wood            3
Understanding
            Explain the chemical treatment and effect of
  M3.04                                                               5
Applying
            chemicals on wood

Contents:
Chemistry of Fibrous Raw Materials
Structure of softwoods, hardwoods, and non-woods; Pulpwood species;
Ultrastructure of
cell wall, cell wall components - cellulose, hemi cellulose, lignin, extractives
and inorganic
components. Difference in cell wall composition of softwood and hard wood
Wood extractives - Essential oils, Tannins, flavanoids and dyestuff, properties
and uses.
Effect of extractives on durability of wood, gluing characteristics, surface
coating. Pectin -
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Illustrate the wood hydrolysis and destructive distillation of wood.

This module is presented from the official syllabus scope for Wood Chemistry. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the wood hydrolysis and destructive distillation of wood.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain wood analysis principles

Explain wood analysis principles is an official topic in Module 4 of Wood Chemistry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain wood analysis principles using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain wood analysis principles should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain wood analysis principles means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Explain wood analysis principles [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe distillation of wood

Describe distillation of wood is an official topic in Module 4 of Wood Chemistry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe distillation of wood using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe distillation of wood should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe distillation of wood means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Describe distillation of wood □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Suggest different types of wood hydrolysis 3 Understanding Wood Analysis. - Preparation of sample - Moisture content-oven dry method and dean and stark Ash content, Cellulose - cross &bevan, alpha, beta, gamma, holocellulose Lignin, Extractives - steam distillation and solvent extraction (soxhlet extraction) Destructive distillation of wood - charcoal, Pyroligneous acid, wood tar. Wood hydrolysis - Acid hydrolysis, Bio - Chemical hydrolysis, enzymatic hydrolysis Text /Reference: T/R Book Title/Author T1 R. H. Farmer, Chemistry in the utilization of wood, Elsevier, 2013 Kocurek M. J., "Pulp and Paper Manufacture, Volume 1: Properties of Fibrous R1 Raw Materials and their Preparation for Pulping (Ed. Kocurek M. J. and Stevens C. F.B.), TAPPI Press, 1983 Franz F.P. Kollmann, Wilfred A.Jr. Cote, Principles of wood science & R2 technology vol. I,Springer Berlin Heidelberg, 2012 R3 Browning B. L. "The Chemistry of Wood", John Wiley & Sons. 1981 Sjostrom E., "Wood Chemistry Fundamentals and Applications", 2nd Ed., R4 TAPPI Press. 1993 K. S.Tewari, S. N Mehrotra, N. K. Vishnoi , "Textbook of organic chemistry" R5 Vikas Publishing House Pvt Ltd., B.R. Puri, L.R. Sharma, M.S. Pathania ,Physical chemistry, Vishal Publishing R6 Company Online Resources: Sl. No Website Link 1 http://www.ncert.nic.in/NCERTS/l/kech205.pdf / 2 http://ncert.nic.in/NCERTS/l/leph205.pdf

Suggest different types of wood hydrolysis 3 Understanding Wood Analysis. - Preparation of sample - Moisture content-oven dry method and dean and stark Ash content, Cellulose - cross &bevan, alpha, beta, gamma, holocellulose Lignin, Extractives - steam distillation and solvent extraction (soxhlet extraction) Destructive distillation of wood - charcoal, Pyroligneous acid, wood tar. Wood hydrolysis - Acid hydrolysis, Bio - Chemical hydrolysis, enzymatic hydrolysis Text /Reference: T/R Book Title/Author T1 R. H. Farmer, Chemistry in the utilization of wood, Elsevier, 2013 Kocurek M. J., "Pulp and Paper Manufacture, Volume 1: Properties of Fibrous R1 Raw Materials and their Preparation for Pulping (Ed. Kocurek M. J. and Stevens C. F.B.), TAPPI Press, 1983 Franz F.P. Kollmann, Wilfred A.Jr. Cote, Principles of wood science & R2 technology vol. I,Springer Berlin Heidelberg, 2012 R3 Browning B. L. "The Chemistry of Wood", John Wiley & Sons. 1981 Sjostrom E., "Wood Chemistry Fundamentals and Applications", 2nd Ed., R4 TAPPI Press. 1993 K. S.Tewari, S. N Mehrotra, N. K. Vishnoi , "Textbook of organic chemistry" R5 Vikas Publishing House Pvt Ltd., B.R. Puri, L.R. Sharma, M.S. Pathania ,Physical chemistry, Vishal Publishing R6 Company Online Resources: Sl. No Website Link 1 http://www.ncert.nic.in/NCERTS/l/kech205.pdf / 2 http://ncert.nic.in/NCERTS/l/leph205.pdf is an official topic in Module 4 of Wood Chemistry. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Suggest different types of wood hydrolysis 3 Understanding Wood Analysis. - Preparation of sample - Moisture content-oven dry method and dean and stark Ash content, Cellulose - cross &bevan, alpha, beta, gamma, holocellulose Lignin, Extractives - steam distillation and solvent extraction (soxhlet extraction) Destructive distillation of wood - charcoal, Pyroligneous acid, wood tar. Wood hydrolysis - Acid hydrolysis, Bio - Chemical hydrolysis, enzymatic hydrolysis Text /Reference: T/R Book Title/Author T1 R. H. Farmer, Chemistry in the utilization of wood, Elsevier, 2013 Kocurek M. J., “Pulp and Paper Manufacture, Volume 1: Properties of Fibrous R1 Raw Materials and their Preparation for Pulping (Ed. Kocurek M. J. and Stevens C. F.B.), TAPPI Press, 1983 Franz F.P. Kollmann, Wilfred A.Jr. Cote, Principles of wood science & R2 technology vol. I,Springer Berlin Heidelberg, 2012 R3 Browning B. L. “The Chemistry of Wood”, John Wiley & Sons. 1981 Sjostrom E., “Wood Chemistry Fundamentals and Applications”, 2nd Ed., R4 TAPPI Press. 1993 K. S.Tewari, S. N Mehrotra, N. K. Vishnoi , “Textbook of organic chemistry” R5 Vikas Publishing House Pvt Ltd., B.R. Puri, L.R. Sharma, M.S. Pathania ,Physical chemistry, Vishal Publishing R6 Company Online Resources: Sl. No Website Link 1 http://www.ncert.nic.in/NCERTS/l/kech205.pdf / 2 http://ncert.nic.in/NCERTS/l/leph205.pdf using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, suggest different types of wood hydrolysis 3 understanding wood analysis. - preparation of sample - moisture content-oven dry method and dean and stark ash content, cellulose - cross &bevan, alpha, beta, gamma, holocellulose lignin, extractives - steam distillation and solvent extraction (soxhlet extraction) destructive distillation of wood - charcoal, pyroligneous acid, wood tar. wood hydrolysis - acid hydrolysis, bio - chemical hydrolysis, enzymatic hydrolysis text /reference: t/r book title/author t1 r. h. farmer, chemistry in the utilization of wood, elsevier, 2013 kocurek m. j., “pulp and paper manufacture, volume 1: properties of fibrous r1 raw materials and their preparation for pulping (ed. kocurek m. j. and stevens c. f.b.), tappi press, 1983 franz f.p. kollmann, wilfred a.jr. cote, principles of wood science & r2 technology vol. i,springer berlin heidelberg, 2012 r3 browning b. l. “the chemistry of wood”, john wiley & sons. 1981 sjostrom e., “wood chemistry fundamentals and applications”, 2nd ed., r4 tappi press. 1993 k. s.tewari, s. n mehrotra, n. k. vishnoi , “textbook of organic chemistry” r5 vikas publishing house pvt ltd., b.r. puri, l.r. sharma, m.s. pathania ,physical chemistry, vishal publishing r6 company online resources: sl. no website link 1 http://www.ncert.nic.in/ncerts/l/kech205.pdf / 2 http://ncert.nic.in/ncerts/l/leph205.pdf should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Suggest different types of wood hydrolysis 3 Understanding Wood Analysis. - Preparation of sample - Moisture content-oven dry method and dean and stark Ash content, Cellulose - cross &bevan, alpha, beta, gamma, holocellulose Lignin, Extractives - steam distillation and solvent extraction (soxhlet extraction) Destructive distillation of wood - charcoal, Pyroligneous acid, wood tar. Wood hydrolysis - Acid hydrolysis, Bio - Chemical hydrolysis, enzymatic hydrolysis Text /Reference: T/R Book Title/Author T1 R. H. Farmer, Chemistry in the utilization of wood, Elsevier, 2013 Kocurek M. J., “Pulp and Paper Manufacture, Volume 1: Properties of Fibrous R1 Raw Materials and their Preparation for Pulping (Ed. Kocurek M. J. and Stevens C. F.B.), TAPPI Press, 1983 Franz F.P. Kollmann, Wilfred A.Jr. Cote, Principles of wood science & R2 technology vol. I,Springer Berlin Heidelberg, 2012 R3 Browning B. L. “The Chemistry of Wood”, John Wiley & Sons. 1981 Sjostrom E., “Wood Chemistry Fundamentals and Applications”, 2nd Ed., R4 TAPPI Press. 1993 K. S.Tewari, S. N Mehrotra, N. K. Vishnoi , “Textbook of organic chemistry” R5 Vikas Publishing House Pvt Ltd., B.R. Puri, L.R. Sharma, M.S. Pathania ,Physical chemistry, Vishal Publishing R6 Company Online Resources: Sl. No Website Link 1 http://www.ncert.nic.in/NCERTS/l/kech205.pdf / 2 http://ncert.nic.in/NCERTS/l/leph205.pdf means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Suggest different types of wood hydrolysis 3 Understanding Wood Analysis. - Preparation of sample - Moisture content-oven dry method and dean and stark Ash content, Cellulose - cross &bevan, alpha, beta, gamma, holocellulose Lignin, Extractives - steam distillation and solvent extraction (soxhlet extraction) Destructive distillation of wood - charcoal, Pyroligneous acid, wood tar. Wood hydrolysis - Acid hydrolysis, Bio - Chemical hydrolysis, enzymatic hydrolysis Text /Reference: T/R Book Title/Author T1 R. H. Farmer, Chemistry in the utilization of wood, Elsevier, 2013 Kocurek M. J., “Pulp and Paper Manufacture, Volume 1: Properties of Fibrous R1 Raw Materials and their Preparation for Pulping (Ed. Kocurek M. J. and Stevens C. F.B.), TAPPI Press, 1983 Franz F.P. Kollmann, Wilfred A.Jr. Cote, Principles of wood science & R2 technology vol. I,Springer Berlin Heidelberg, 2012 R3 Browning B. L. “The Chemistry of Wood”, John Wiley & Sons. 1981 Sjostrom E., “Wood Chemistry Fundamentals and Applications”, 2nd Ed., R4 TAPPI Press. 1993 K. S.Tewari, S. N Mehrotra, N. K. Vishnoi , “Textbook of organic chemistry” R5 Vikas Publishing House Pvt Ltd., B.R. Puri, L.R. Sharma, M.S. Pathania ,Physical chemistry, Vishal Publishing R6 Company Online Resources: Sl. No Website Link 1 http://www.ncert.nic.in/NCERTS/l/kech205.pdf / 2 http://ncert.nic.in/NCERTS/l/leph205.pdf □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Illustrate the wood hydrolysis and destructive distillation of wood.

  M4.01     Explain wood analysis principles                          3
Understanding

  M4.02     Describe distillation of wood                             3
Understanding

  M4.03     Suggest different types of wood hydrolysis                3
Understanding

            Series Test – II                                          1

Contents:
Wood Analysis. - Preparation of sample - Moisture content-oven dry method and
dean and
stark Ash content, Cellulose - cross &bevan, alpha, beta, gamma, holocellulose
Lignin, Extractives - steam distillation and solvent extraction ( soxhlet
extraction)

Destructive distillation of wood - charcoal, Pyroligneous acid, wood tar.
Wood hydrolysis - Acid hydrolysis, Bio - Chemical hydrolysis, enzymatic
hydrolysis

Text /Reference:

  T/R                                          Book Title/Author
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
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Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 3 Understanding nomenclature Discuss the preparation, properties and uses of. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding nomenclature Discuss the preparation, properties and uses of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 2 Understanding organic compounds Explain the structure of cellulose, glucose and. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding organic compounds Explain the structure of cellulose, glucose and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Understanding fructose List wood extractive terpenes and their isolation. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding fructose List wood extractive terpenes and their isolation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 2 Understanding from plants Organic Chemistry Classification of organic compounds - saturated and unsaturated compounds. IUPAC - nomenclature of simple organic compounds. Industrial methods of preparation, properties and uses of : Alcohol, phenol, aldehydes, ketones, carboxyllic acid, amines, urea, poly - functional compounds - glycol, glycerol, Carbohydrates - classification of sugars, structure of glucose, fructose and cellulose, important reaction of these compounds. Terpenes - Essential oils, classification of terpenoids, Isolation, Isoprene rule and the examples of terpenes.. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding from plants Organic Chemistry Classification of organic compounds - saturated and unsaturated compounds. IUPAC - nomenclature of simple organic compounds. Industrial methods of preparation, properties and uses of : Alcohol, phenol, aldehydes, ketones, carboxyllic acid, amines, urea, poly - functional compounds - glycol, glycerol, Carbohydrates - classification of sugars, structure of glucose, fructose and cellulose, important reaction of these compounds. Terpenes - Essential oils, classification of terpenoids, Isolation, Isoprene rule and the examples of terpenes.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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Module 2

Define or explain 3 Understanding corrosion on wood Analyse the colloids and their importance in. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding corrosion on wood Analyse the colloids and their importance in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Applying paper making chemistry. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying paper making chemistry*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Apply the carbon dating to find the age of wood 3 Applying Physical Chemistry Corrosion-Theory of corrosion-dry & wet, methods to prevent corrosion. Degradation of wood by products of metal corrosion. Introduction to colloids, mechanical, electrical and optical properties of colloids. Gels & Emulsions; Importance of papermaking chemistry; Fiber-fiber water bonding; Rheology, surface energy, and surface tension of colloidal systems. Electro kinetic properties, Sorption and swelling of cellulosic materials. Nuclear Chemistry Natural and artificial radioactivity, radioisotopes and their uses, carbon dating in wood Discuss the structure, cell wall composition of wood and its chemical. (3 marks)

Answer: Begin with the standard definition or identity of *Apply the carbon dating to find the age of wood 3 Applying Physical Chemistry Corrosion-Theory of corrosion-dry & wet, methods to prevent corrosion. Degradation of wood by products of metal corrosion. Introduction to colloids, mechanical, electrical and optical properties of colloids. Gels & Emulsions; Importance of papermaking chemistry; Fiber-fiber water bonding; Rheology, surface energy, and surface tension of colloidal systems. Electro kinetic properties, Sorption and swelling of cellulosic materials. Nuclear Chemistry Natural and artificial radioactivity, radioisotopes and their uses, carbon dating in wood Discuss the structure, cell wall composition of wood and its chemical*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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Module 3

Define or explain 3 Understanding between soft wood and hardwood Explain cell wall structure and composition of. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding between soft wood and hardwood Explain cell wall structure and composition of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding wood. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding wood*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Summarize extractives and their effect on wood 3 Understanding Explain the chemical treatment and effect of. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize extractives and their effect on wood 3 Understanding Explain the chemical treatment and effect of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Applying chemicals on wood Chemistry of Fibrous Raw Materials Structure of softwoods, hardwoods, and non-woods; Pulpwood species; Ultrastructure of cell wall, cell wall components - cellulose, hemi cellulose, lignin, extractives and inorganic components. Difference in cell wall composition of softwood and hard wood Wood extractives - Essential oils, Tannins, flavanoids and dyestuff, properties and uses. Effect of extractives on durability of wood, gluing characteristics, surface coating. Pectin - uses

Chemical Treatment of wood - chemical resistance of softwood and hardwood. Effect of chemicals on wood - aqueous solutions of inorganic salts, alkaline degradation of wood, Oxidation and bleaching of wood. Action of water - effect of moisture content on strength & electrical conductivity of wood. (3 marks)

Answer: Begin with the standard definition or identity of *5 Applying chemicals on wood Chemistry of Fibrous Raw Materials Structure of softwoods, hardwoods, and non-woods; Pulpwood species; Ultrastructure of cell wall, cell wall components - cellulose, hemi cellulose, lignin, extractives and inorganic components. Difference in cell wall composition of softwood and hard wood Wood extractives - Essential oils, Tannins, flavanoids and dyestuff, properties and uses. Effect of extractives on durability of wood, gluing characteristics, surface coating. Pectin - uses Chemical Treatment of wood - chemical resistance of softwood and hardwood. Effect of chemicals on wood - aqueous solutions of inorganic salts, alkaline degradation of wood, Oxidation and bleaching of wood. Action of water - effect of moisture content on strength & electrical conductivity of wood*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Explain wood analysis principles. (3 marks)

Answer: Begin with the standard definition or identity of *Explain wood analysis principles*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe distillation of wood. (3 marks)

Answer: Begin with the standard definition or identity of *Describe distillation of wood*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Suggest different types of wood hydrolysis 3 Understanding Wood Analysis. - Preparation of sample - Moisture content-oven dry method and dean and stark Ash content, Cellulose - cross &bevan, alpha, beta, gamma, holocellulose Lignin, Extractives - steam distillation and solvent extraction (soxhlet extraction) Destructive distillation of wood - charcoal, Pyroligneous acid, wood tar. Wood hydrolysis - Acid hydrolysis, Bio - Chemical hydrolysis, enzymatic hydrolysis Text /Reference: T/R Book Title/Author T1 R. H. Farmer, Chemistry in the utilization of wood, Elsevier, 2013 Kocurek M. J., “Pulp and Paper Manufacture, Volume 1: Properties of Fibrous R1 Raw Materials and their Preparation for Pulping (Ed. Kocurek M. J. and Stevens C. F.B.), TAPPI Press, 1983 Franz F.P. Kollmann, Wilfred A.Jr. Cote, Principles of wood science & R2 technology vol. I,Springer Berlin Heidelberg, 2012 R3 Browning B. L. “The Chemistry of Wood”, John Wiley & Sons. 1981 Sjostrom E., “Wood Chemistry Fundamentals and Applications”, 2nd Ed., R4 TAPPI Press. 1993 K. S.Tewari, S. N Mehrotra, N. K. Vishnoi , “Textbook of organic chemistry” R5 Vikas Publishing House Pvt Ltd., B.R. Puri, L.R. Sharma, M.S. Pathania ,Physical chemistry, Vishal Publishing R6 Company Online Resources: Sl. No Website Link 1 http://www.ncert.nic.in/NCERTS/l/kech205.pdf / 2 http://ncert.nic.in/NCERTS/l/leph205.pdf. (3 marks)

Answer: Begin with the standard definition or identity of *Suggest different types of wood hydrolysis 3 Understanding Wood Analysis. - Preparation of sample - Moisture content-oven dry method and dean and stark Ash content, Cellulose - cross &bevan, alpha, beta, gamma, holocellulose Lignin, Extractives - steam distillation and solvent extraction (soxhlet extraction) Destructive distillation of wood - charcoal, Pyroligneous acid, wood tar. Wood hydrolysis - Acid hydrolysis, Bio - Chemical hydrolysis, enzymatic hydrolysis Text /Reference: T/R Book Title/Author T1 R. H. Farmer, Chemistry in the utilization of wood, Elsevier, 2013 Kocurek M. J., “Pulp and Paper Manufacture, Volume 1: Properties of Fibrous R1 Raw Materials and their Preparation for Pulping (Ed. Kocurek M. J. and Stevens C. F.B.), TAPPI Press, 1983 Franz F.P. Kollmann, Wilfred A.Jr. Cote, Principles of wood science & R2 technology vol. I,Springer Berlin Heidelberg, 2012 R3 Browning B. L. “The Chemistry of Wood”, John Wiley & Sons. 1981 Sjostrom E., “Wood Chemistry Fundamentals and Applications”, 2nd Ed., R4 TAPPI Press. 1993 K. S.Tewari, S. N Mehrotra, N. K. Vishnoi , “Textbook of organic chemistry” R5 Vikas Publishing House Pvt Ltd., B.R. Puri, L.R. Sharma, M.S. Pathania ,Physical chemistry, Vishal Publishing R6 Company Online Resources: Sl. No Website Link 1* *http://www.ncert.nic.in/NCERTS/l/kech205.pdf / 2 http://ncert.nic.in/NCERTS/l/leph205.pdf*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Understanding nomenclature Discuss the preparation, properties and uses of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **3 Understanding corrosion on wood Analyse the colloids and their importance in**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **3 Understanding between soft wood and hardwood Explain cell wall structure and composition of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain wood analysis principles**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No Website Link
- http://www.ncert.nic.in/NCERTS/l/kech205.pdf / http://ncert.nic.in/NCERTS/l/leph205.pdf

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